MADHUKAR REDDY MUSKU

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EDUCATION

University of Iowa, Iowa City, IA
Master of Science in Data Science | August 2023 – May 2025 | CGPA: **3.80 / 4.00**

National Institute of Technology (NIT) Tiruchirappalli, India
Bachelor of Technology in Engineering | August 2019 – May 2023 | CGPA: **3.50 / 4.00**

Delta Junior College, Hyderabad, India
Intermediate (Mathematics, Physics, Computer Science) | May 2018 | CGPA: **3.80 / 4.00**

Vijetha High School, Hyderabad, India
Secondary School Certificate (Mathematics, Physics, Chemistry) | May 2016 | CGPA: **4.00 / 4.00**

COURSEWORK

Probability and Statistics; Statistical Learning; Probabilistic Statistical Learning; Computing in Statistics; Applied Linear Regression; Data Visualization and Data Technologies; Fundamentals of Software Engineering; Software Engineering Languages and Tools; Software Architectures for Health Care Information Systems; Generative Artificial Intelligence Tools; MS Data Science Practicum.

EXPERIENCE

Data Analyst / Research Data Scientist – University of Iowa Health Care ***June 2025 – Present***

- Lead end-to-end development, deployment, and maintenance of the PTAGQ platform, owning frontend (React), backend (Flask/Node), relational databases (PostgreSQL/MySQL), cloud infrastructure (AWS), security, and automated testing.
- Maintain and extend the PTGQ platform, implementing new features, system upgrades, performance improvements, and data integrity checks.
- Design database schemas, access controls, and authentication workflows to support secure, scalable multi-institutional data collection.
- Manage cloud resources, CI/CD pipelines, role-based access, and compliance workflows to ensure reliability, security, and reproducibility.
- Collaborate with faculty and national stakeholders to translate research requirements into production-ready software systems.

Research Intern (ML Engineer) ***Nov 2024-May 2025***
McArdle Laboratory for Cancer Research, University of Wisconsin–Madison

- Developed deep learning models (CNNs, 3D U-Net) for automated cell-cycle phase classification (G1/S/G2/M) from 3D fluorescence microscopy stacks, achieving a 23% F1-score improvement over baselines.
- Designed a robust 3D instance segmentation pipeline combining watershed algorithms and custom post-processing to detect multi-cell structures across z-stacks, substantially reducing manual annotation effort.
- Conducted large-scale hyperparameter optimization (120+ experiments) with MLflow and distributed GPU training to ensure reproducible and scalable volumetric imaging research.
- Built an end-to-end research platform (React + Flask) enabling batch upload, automated segmentation, classification, and human-in-the-loop validation, reducing analysis time from hours to minutes.
- Collaborated closely with PI and experimental biologists to address domain shift in real microscopy data, improving model robustness across varying staining and noise conditions

Graduate Research Assistant, Physical Therapy & Rehabilitation Science, UIowa ***Aug 2024-May 2025***

- Designed and maintained scalable ETL pipelines to ingest, clean, validate, and transform large multi-institutional PT education survey datasets.
- Applied advanced statistical analysis and machine learning (regression, CART, classification) to identify predictors of learner preparedness, burnout, empathy, uncertainty tolerance, and educational debt.
- Built automated benchmarking, reporting, and longitudinal analytics used for APTA accreditation, certification review, and national program ranking.
- Developed interactive dashboards and communicated analytical findings to faculty and national education stakeholders; contributed to trademarked educational analytics tools.
- Built automated time-series pipelines to analyze and visualize CPI competency progression at individual and

cohort levels, generating longitudinal reports to support academic review and clinical preparedness decisions.

Graduate Research Assistant – Molecular Physiology & Biophysics, UIowa ***June 2024-Aug 2025***

- Analyzed protein–protein interaction (PPI) networks using Cytoscape to identify key molecular pathways and interaction structures.
- Applied graph-based clustering and centrality measures to detect hub proteins and high-importance nodes within biological networks.
- Utilized Graph Neural Networks (GNNs) to assess functional significance and regulatory roles of proteins.
- Employed scGPT and literature-driven data analysis to support pathway-level interpretation of molecular mechanisms.

Graduate Research Assistant, Medical Computer Vision and Graphics Lab, UIowa ***Oct 2023-June 2024***

- Led synthesis of head-and-neck tumor segmentation masks by fusing expert manual annotations with JEI-based consensus ground truth generation.
- Engineered multiple ground-truth variants to study annotation uncertainty and its impact on model learning, using MATLAB, Python, 3D Slicer, and MONAI.
- Trained and evaluated deep learning segmentation models across different GT flavors to assess robustness, generalization, and sensitivity.
- Performed systematic validation and comparative analysis to inform best practices for ground-truth construction in medical image segmentation studies.

Research Assistant – National Institute of Technology (NIT), Tiruchirappalli ***Jan 2022-May 2023***

- Conducted applied machine learning and statistical modeling for industrial predictive maintenance and environmental analytics using CNNs, LSTMs, and classical ML methods.
- Built deep learning models integrated with IoT sensor data to predict equipment failures, applying ensemble learning and Bayesian hyperparameter optimization to improve reliability and reduce downtime.
- Performed advanced statistical analysis (Four-way ANOVA, multiple linear regression) on indoor air quality data to study environmental and behavioral health impacts.
- Developed and evaluated predictive models for retail price discount forecasting using decision trees, random forests, and boosting techniques**.**

PROJECTS

Multi-GAIT Teaching Assistant Chatbot – AI-Based Learning Companion for PTRS:6224

LLMs, RAG, Python (Flask), ReactJS, Tailwind, Gen AI (OpenAI, Stable Diffusion), Google TTS

- Demonstrated the ability to embed scientific publications (PDFs) and lecture videos (MP4 with transcripts) into an AI-powered course support system, enhancing student access to complex topics in muscle mechanics, genomics, epigenetics, and neurophysiology.
- Built a Retrieval-Augmented Generation (RAG) pipeline that processed and indexed lecture content and publications for contextual response generation via GPT-3.5-turbo.
- Developed a React-based multi-modal frontend enabling students to interact with course material through text explanations, audio answers, and AI-generated anatomical illustrations.
- Achieved 24/7 scalable educational support with sub-30-second response time, improving learner engagement outside class hours.

Automated Nasogastric (NG) Tube Detection and Classification from Abdominal X-rays

Deep Learning, Medical Imaging, Computer Vision, PyTorch, Grad-CAM

- Developed an end-to-end deep learning pipeline for nasogastric (NG) tube detection and placement classification from abdominal X-ray images.
- Implemented U-Net–based segmentation to isolate NG tubes, followed by CNN-based classification to distinguish correct vs. incorrect placement.
- Worked with ~600 labeled X-ray images (segmentation masks + classification labels), applying data augmentation and model evaluation using AUC and accuracy metrics.
- Integrated explainability (Grad-CAM) and optimized training using pretrained medical vision models to improve robustness and clinical relevance.

Virtual Disaster Assistance Platform Development (Double Iterative Hybrid Agile Process)

Java, Angular, MySQL, Agile, REST APIs, Automated Testing

- Developed Virtual Disaster Assistance Platform using a Double Iterative Hybrid Agile process, integrating automated request-response matching for global disaster relief.

- Utilized Java, Angular, and MySQL for backend and frontend development, ensuring a secure, user-friendly experience with role-based access and data encryption.
- Applied GitHub, jUnit, Selenium, and SonarQube for version control, automated testing, and continuous code quality monitoring, supporting efficient, collaborative development and deployment.

Integrated Healthcare Management System (IHMS): Layered Architecture and FHIR Implementation

Microservices, REST APIs, FHIR, Java, Node.js, MySQL, HIPAA Compliance

- Spearheaded Layered Architecture implementation for an Integrated Healthcare Management System, focusing on EHR and patient management.
- Engineered secure web solutions with Java, .NET, Python, and Node.js, leveraging microservices and RESTful APIs for enhanced system interoperability.
- Integrated FHIR standards for seamless data exchange; utilized MySQL for data management ensuring compliance with HIPAA and robust security via RBAC and 128-bit encryption.
- Streamlined patient care processes and improved operational efficiency through a collaborative development approach and user-centric design.

Multiplayer RPG Board Game with Generative AI (Ruby on Rails)

Ruby on Rails, PostgreSQL, REST APIs, GenAI (LLM Integration), Game Logic, MVC Architecture

- Designed and developed a multiplayer role-playing board game in Ruby on Rails where players collaboratively and competitively explore a grid-based world to discover hidden treasure.
- Implemented core game mechanics including turn-based movement, player roles, scoring, game state persistence, and win conditions using Rails MVC architecture.
- Integrated Generative AI (LLM) to dynamically generate in-game narratives, quests, hints, and adaptive gameplay elements based on player actions, enhancing replayability and engagement.
- Built secure user authentication, session management, and multiplayer synchronization using RESTful endpoints and database-backed game states.
- Applied software engineering best practices including modular design, RESTful routing, database normalization, and automated testing for game logic validation.

Innovative Predictive Modeling for Chronic Hepatitis B

Machine Learning, NHANES, Feature Engineering, Model Evaluation, Healthcare Analytics

- Developed a highly targeted predictive model for Chronic Hepatitis B (HBV) infection risk among individuals with unclear vaccination history, leveraging NHANES dataset insights.
- Crafted a model focusing on unvaccinated or vaccination-uncertain populations, meticulously filtering out previously diagnosed cases to boost the model's clinical relevance.
- Integrated a blend of demographic and medical data indicators to identify high-risk groups, guiding efficient and directed blood testing protocols.
- Emphasized model reliability and ethical data use, ensuring variables were practical, routinely collected in medical settings, and sensitive to patient disclosure challenges regarding risk behaviors and socioeconomic factors.

TECHNICAL SKILLS

Programming Languages: Python (primary), R, SQL, MATLAB, Ruby, JavaScript, Bash, Java, C++

Machine Learning & AI: Deep Learning, Statistical Learning, Computer Vision, Time Series Analysis, Graph-based Learning, CNNs, RNNs/LSTMs, Transformers, GNNs, Model Evaluation & Validation

Frameworks & Libraries: PyTorch, TensorFlow, Keras, scikit-learn, MONAI, OpenCV, SimpleITK, LangChain, HuggingFace Transformers

Data Engineering & Databases: ETL Pipeline Design, Data Cleaning & Validation, Reproducible Analytics,

PostgreSQL, MySQL, MongoDB, Spark (basic), Hadoop (basic)

Web & Systems Development: React, Flask, Node.js, Express, Ruby on Rails, RESTful APIs, Authentication & Authorization, Automated Testing

Cloud & DevOps: AWS (EC2, RDS, S3, IAM), Linux/Unix, Git, CI/CD, Containerized Workflows (Docker – basic)

Scientific & Biomedical Computing: 3D Slicer, Cytoscape, Fiji/ImageJ, Medical Image Segmentation, Proteomics Network Analysis

Visualization & Reporting: Matplotlib, Seaborn, Tableau, Power BI, Interactive Dashboards, Analytics